

# Kyle Solowiej Hawkins

1668 N Forgeus Ave, Tucson, AZ, 85716  
(303) 518-2637, [ksolohawk@gmail.com](mailto:ksolohawk@gmail.com)  
Website: [ksolohawk.com](http://ksolohawk.com)

## Work Experience

- Airy Optics Inc.** - Senior Optical Engineer **September 2022 - Present** (Tucson, AZ)
- Developed polarization based scattering algorithms, integrated Scatmech routines into a ray trace
  - Biaxial and Uniaxial coating modeling and analysis
  - Pancake lens/VR system model set up and analysis, and polarized stray light modeling
- Arete Associates** - Optical Systems Engineer (II) **August 2018 – September 2022** (Tucson, AZ)
- Assembled and tested multispectral imaging systems (MSI) for the COBRA program
  - Prototyped a multispectral stokes imaging polarimeter - designed filters and image processing routines to extract the stokes parameters from raw image data
  - CMOS sensor alignment, testing and characterization
- Airy Optics Inc.** - Algorithm Scientist **May 2016 – August 2018** (Tucson, AZ)
- Developed and image simulation software to model the effect of dichroic dyes on polarized images
  - Modeled/analyzed optical systems during engineering service projects with Polaris-M, interferometers, compound retarders, depolarizers etc.
  - Taught classes to Polaris-M customers on using the software for optical analysis
- Zemax LLC.** - Optical Engineering Intern **June 2015 - August 2015** (Kirkland, WA)
- Wrote Macros (high-level set of commands to retrieve data not available in the Zemax GUI) for tolerancing extended aspheric surfaces
  - Performed study on the optical performance of Extended Aspheric during optimization • Authored informative articles on Polarization for the Zemax Knowledge Base ([see my article](#))
- University of Arizona** - Undergraduate Research Assistant/U of A Think Tank **August 2013 – May 2016** (Tucson, AZ)
- Updated web server (Node.js) for remote access to optical polarization ray tracing software
  - Created computer generated graphics to describe polarization phenomena
  - Led supplemental review sessions for the Electricity and Magnetism course PHYS 241
  - Tutored students in calculus I, calculus II and algebra math classes - Certified Level 1 Tutor

## Education

**Master of Science:** University of Arizona, Wyant College of Optical Sciences (specializing in biomedical optics)

See my published thesis here: [Depolarization of Biological Tissues...](#)

**Institution:** University of Arizona, Honors College (August 2012- May 2016)

**Dual Major:** Optical Sciences and Engineering (B.S.), Applied Mathematics (B.S.), University of Arizona, Honors College

## Technical Experience

**Programming Languages:** Python, MATLAB, Mathematica, JavaScript

**Optical Analysis Software:** Zemax, CodeV, Polaris-M

**Development Tools:** Git, GitHub, SVN, Eclipse ([my github](#))

I also love Rock Climbing, playing the Ukulele (and trying to sing) and learning Spanish (es muy divertido!).